Assignment 15

Hands-on Subnetting Practical

1. **Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.**

- New CIDR: /26
- Subnet mask: 255.255.255.192
- Block size: 64

Subnet	Network Address	Broadcast Address
1	192.168.10.0	192.168.10.63
2	192.168.10.64	192.168.10.127
3	192.168.10.128	192.168.10.191
4	192.168.10.192	192.168.10.255

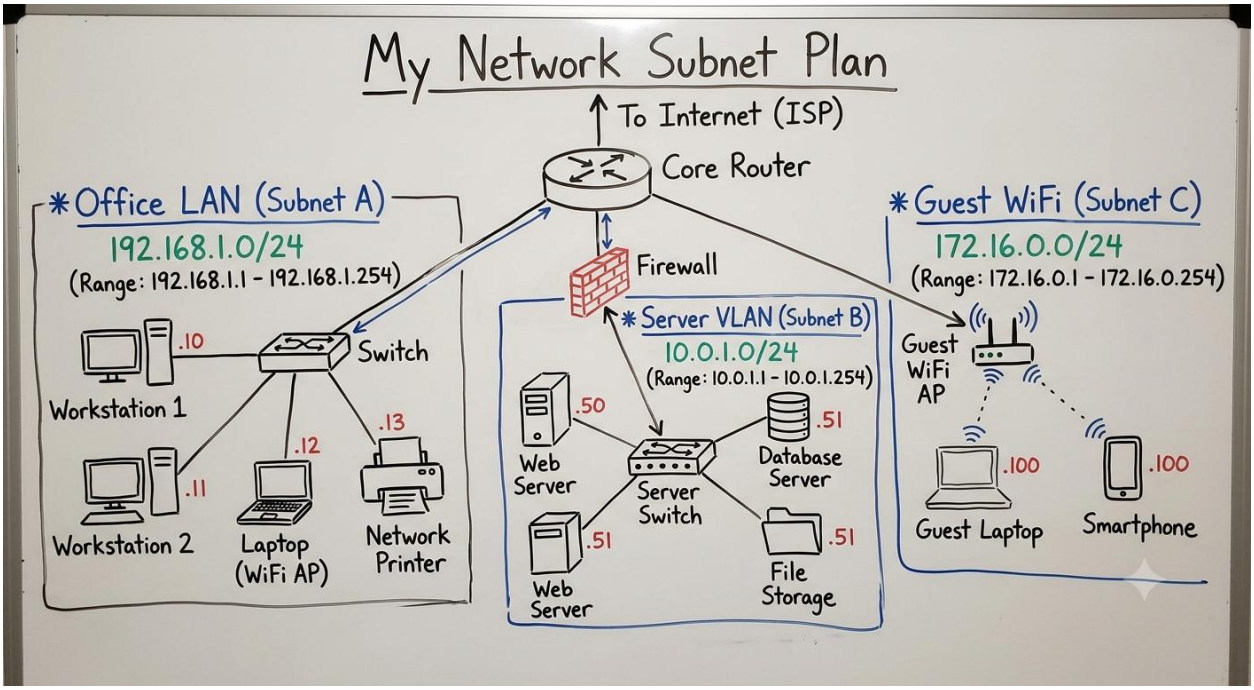
2. **For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device. Hint: Make sure not to use the network or broadcast address for any device.**

Using Subnet 1: 192.168.10.0/26

- Subnet mask: 255.255.255.192
- Network address: 192.168.10.0
- Broadcast address: 192.168.10.63
- Valid host range: 192.168.10.1 to 192.168.10.62

Device	Assigned IP address
Router	192.168.10.1
Laptop 1	192.168.10.2
Laptop 2	192.168.10.3
Printer	192.168.10.4
Mobile	192.168.10.5

3. **Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.**



4. **Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.**

❖ **Given network: 192.168.10.0/24**

Step 1: Calculate the new CIDR

The original network is 24. To create 4 subnets, we need 2 borrowed bits because (2^2 = 4).

New CIDR = /24 + 2 = /26

Step 2: Calculate the subnet mask

The subnet mask for /26 is:

255.255.255.192

Step 3: Calculate the block size

Use the formula:

Block size = 256 − subnet mask value in the interesting octet

Block size = 256 − 192 = 64

Step 4: Find the four subnets

Starting from 0, add the block size of 64 each time:

- Subnet 1: 0 to 63
- Subnet 2: 64 to 127
- Subnet 3: 128 to 191
- Subnet 4: 192 to 255

The complete subnet ranges are:

Subnet	Network address	Broadcast address
1	192.168.10.0	192.168.10.63
2	192.168.10.64	192.168.10.127
3	192.168.10.128	192.168.10.191
4	192.168.10.192	192.168.10.255